

Pause with You

A focused narrative review of sensorimotor ritual, smoking cues, visible exhalation feedback, and design implications for a multisensory smoking-pause device



Portfolio research document • Literature-informed speculative design • September 2026

Abstract

Pause with You is a speculative handheld device that asks whether selected sensorimotor and ritual functions of a smoking break can be reinterpreted without tobacco combustion or nicotine delivery. The concept begins with an autobiographical observation: seeing an exhaled plume appeared to contribute to the designer's feeling of pause and relaxation. That observation is not treated as population evidence. This focused narrative review instead examines four relevant bodies of literature: smoking beyond pharmacology; conditioned and multisensory cue reactivity; respiratory practices and visible feedback; and water-aerosol, hygiene, and claim-safety considerations. Evidence suggests that handling, sensory cues, and non-nicotine smoking-related stimuli can influence craving and behavior, while the same resemblance can also activate conditioned responses. This tension supports a design strategy of preserving functions while transforming form. The proposed final architecture uses a closed-fist, microphone-like grip, a circular oral interface, ordinary ambient air for inhalation, a physically separated one-way exhalation channel, downstream water-mist generation at the rear of the hand, and an isolated near-nose dry-aroma module. The review does not establish cessation efficacy or inhalation safety. It translates evidence and uncertainty into product requirements, safety constraints, and a future validation agenda.

1. Scope and review method

This document is a focused narrative review, not a systematic review or clinical evaluation. Sources were selected because they directly inform the interaction problem: official cessation guidance; controlled studies of denicotinized cigarettes or non-nicotine sensorimotor substitutes; reviews and experiments on smoking cue reactivity; evidence on breathing-based regulation; and public-health or peer-reviewed sources on portable water aerosol. The purpose is design synthesis rather than pooled effect estimation.

Four evidence statuses are kept separate throughout. Published evidence describes findings reported by external sources. Personal observation refers only to the designer's own experience. Design inference translates evidence into a proposed feature but remains contestable. An unvalidated hypothesis identifies something the concept would need to test before making behavioral, safety, or cessation claims.

2. Smoking is more than nicotine delivery

Nicotine dependence, withdrawal, and reinforcement are central to continued smoking, and evidence-based cessation care therefore includes behavioral support and approved pharmacotherapies. A physical ritual device should not obscure that reality. At the same time, smoking is also a repeated interaction composed of retrieval, handling, hand-to-mouth movement, oral draw, inhalation and exhalation, sensory change, visual completion, and return to the prior activity. Contexts such as stress, meals, boredom, work transitions, and social situations may become linked to this sequence.

Experimental work with denicotinized cigarettes and non-nicotine inhalator substitutes indicates that sensorimotor stimulation can affect short-term craving or smoking behavior even without conventional nicotine delivery. Caponnetto and colleagues found that a nicotine-free inhalator was particularly useful for a subgroup

with strong behavioral dependence on cigarette handling. Donny and colleagues reported that smoking without nicotine can still produce reinforcing and behavioral effects. These findings do not prove that any new ritual object will support cessation; they establish that the behavioral layer is worthy of explicit design attention.

3. The familiarity paradox: substitute and cue

Smoking cues can acquire motivational significance through repeated pairing with nicotine and context. Reviews of cue-exposure research and studies using visual, olfactory, and multisensory stimuli show that smoking-related cues can evoke craving, attention, and physiological or behavioral responses. Coffee odor and other contextual signals can also interact with smoking expectations. Consequently, making an alternative look and behave exactly like a cigarette may improve immediate familiarity while preserving or intensifying conditioned associations.

This creates the project's central design contradiction. Familiar manipulation may lower the interaction cost of substitution, yet visual or sensory resemblance may become a trigger. Pause with You therefore preserves selected functions—grip, oral engagement, paced cycles, visible ending—while changing the object category. Its thicker short-cylinder body is held like a compact microphone inside a closed fist rather than between two fingers like a cigarette. The visible plume appears behind the fist, not at the lips. This is a design inference from the literature, not evidence that the transformation eliminates cue reactivity.

4. Breath, ritual, and visible feedback

Slow and diaphragmatic breathing practices have been studied across stress and anxiety-related outcomes, although methods, populations, and effect sizes vary. A recent systematic review concluded that breathing interventions can influence stress and mental-health outcomes but also emphasized heterogeneity and the need for clearer protocols. This evidence supports treating breath as a potentially meaningful interaction medium, not claiming that a consumer device provides therapy.

The designer's personal observation is narrower: seeing an exhaled plume disperse seemed to mark completion and relaxation during a smoking pause. Pause with You converts that observation into a testable design hypothesis: immediate, spatially coherent visual feedback synchronized with exhalation may strengthen the perceived beginning, cycle, and ending of a pause. The final concept avoids scoring respiratory performance. Short or irregular exhalations should still receive proportional feedback, because turning breathing into a pass/fail task could undermine agency and increase self-monitoring.

5. Aroma as context—not lung delivery

The concept allows optional coffee, green tea, black tea, milk-tea, fruit, or neutral modules because smell can carry contextual and autobiographical meaning. However, “food grade” does not establish respiratory safety, and pleasant odor is not equivalent to a safe inhalable formulation. Aroma is therefore treated as a dry, replaceable, near-nose module that remains outside the oral airflow, exhalation pathway, water reservoir, and

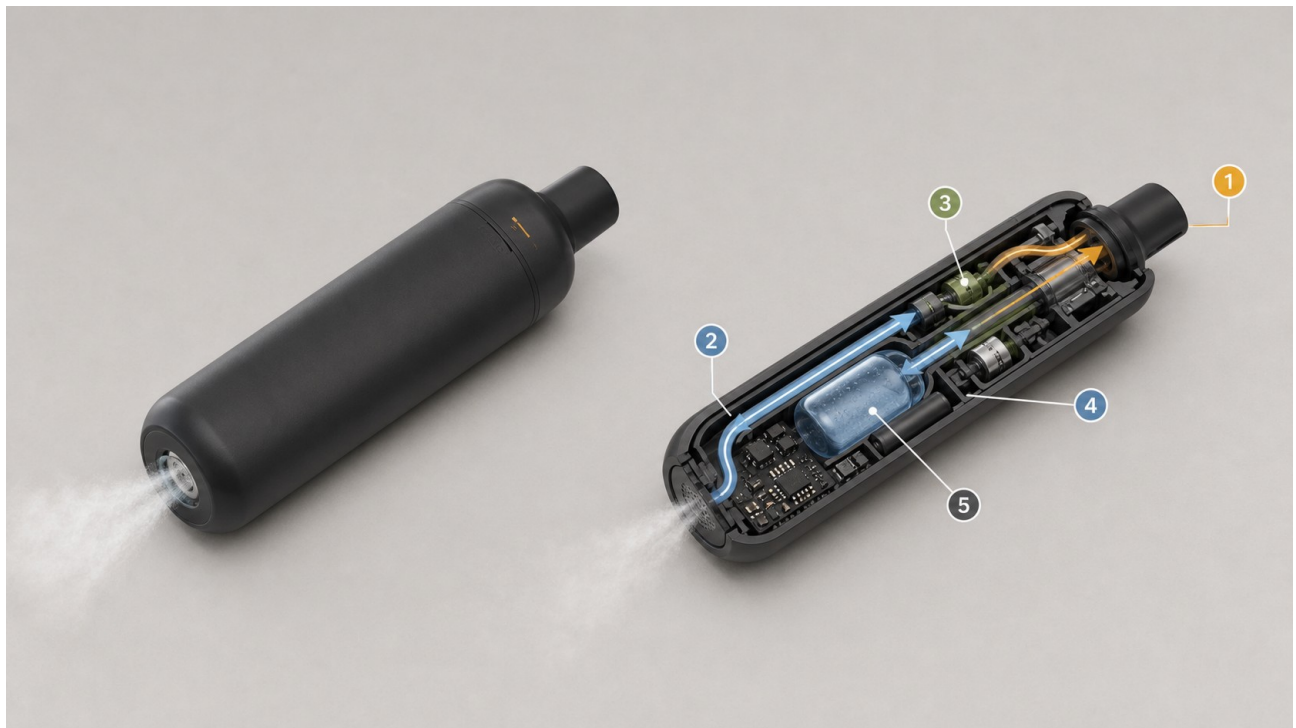
mist generator. The user may choose no aroma. No caffeine, nicotine, active drug, heated liquid, or unknown substance is aerosolized for pulmonary delivery.

This separation also avoids presenting the product as a “healthier electronic cigarette.” Comparative-risk language would require product-specific exposure characterization and appropriate regulatory substantiation. The defensible claim at concept stage is narrower: the intended architecture contains no tobacco combustion and does not intentionally deliver nicotine.

6. Water mist is an exposure, not a neutral visual effect

Ultrasonic and other water-mist systems can disperse minerals, microorganisms, and other constituents present in water or device surfaces. Public-health guidance for humidifiers emphasizes water quality, regular cleaning, drying, and prevention of microbial growth. Peer-reviewed reviews similarly caution that emissions depend on water composition, maintenance, and operating conditions. The phrase “only water” is therefore not a safety argument.

Pause with You uses a route-separation principle. Inhalation is intended to draw ordinary room air through a dry resistive channel. Exhalation passes through a separate one-way path. Only after the exhaled breath has crossed the oral interface and downstream valves does a sensing event activate a separate water-mist generator near the rear of the product. The mist exits behind the closed fist. This architecture is intended to avoid sending generated mist toward the mouth, but it does not establish ambient bystander safety, pulmonary safety, or microbial safety. Those would require engineering verification and exposure testing.



Conceptual architecture: identical exterior shell with physically separated dry-air, exhalation, aroma, water, and control pathways. Internal geometry remains illustrative rather than engineering-ready.

7. Design translation: final Palm-Grip Form

Evidence or constraint				Design translation	Status
Sensorimotor matter	behavior	may		Circular mouthpiece, tactile draw resistance, repeated inhale-exhale cycles	Evidence-informed inference
Smoking cues may trigger craving				Thicker microphone-like power grip; plume exits behind fist	Unvalidated transformation
Breath can structure a pause				Immediate proportional exhalation feedback; no performance score	Evidence-informed inference
Aroma can be a contextual cue				Optional isolated near-nose dry-aroma module	Unvalidated hypothesis
Water contaminants	aerosol	can	carry	Removable reservoir, leak sensing, cleaning logic, rear outlet	Safety requirement
Cessation established	efficacy	is	not	No clinical outcome claim; link to evidence-based support	Ethical boundary

8. Interaction and ergonomic rationale

The final body is a short elongated cylinder approximately 105–125 mm long and 28–34 mm in diameter. It is enclosed in a one-handed power grip similar to holding a compact microphone. The mouthpiece emerges between thumb and index finger; the rear outlet remains beyond the little-finger or wrist side. The user does not remove the product between inhalation and exhalation. This continuity preserves the felt sequence the designer considers important while allowing the visual plume to appear behind the fist.

Human-factors questions remain open: accommodation of hand-size variability; left- and right-hand use; grip force; outlet occlusion; mouthpiece alignment; saliva management; valve opening pressure; cleaning errors; condensation; acoustic and haptic feedback; accessibility for users with limited hand strength; and whether the closed-fist gesture is socially acceptable across contexts. These are requirements for later evaluation, not details that can be resolved by rendering alone.

9. Safety, ethics, and claim boundaries

The concept should not be represented as safe, healthy, therapeutic, or effective for quitting until evidence supports the relevant claim. A responsible portfolio presentation may state the intended absence of combustion and nicotine delivery, identify route separation, and explain foreseeable hazards. It should also acknowledge that mimicking smoking may normalize the gesture, maintain conditioned behavior, appeal to nonsmokers, create shared-air exposure, or delay use of established cessation support.

Minimum concept safeguards include adult-only positioning; no youth-oriented flavors or imagery; no drug delivery; physical separation of aroma and mist; one-way valves; a saliva and condensation trap; leak, temperature, and reservoir-state detection; default shutdown on faults; a removable washable mouthpiece; clear water and cleaning instructions; finite use sessions; and transparent, user-adjustable recommendations. If the product ever moves beyond a portfolio concept, regulatory classification and human-subjects review would need professional assessment in the intended market.

10. Limitations and future validation

No questionnaire, interview, usability study, aerosol characterization, mechanical prototype, or clinical trial was conducted for this project. The images are concept renderings. The literature search was focused and narrative rather than exhaustive. Evidence on existing cigarettes, denicotinized cigarettes, inhalators, cue exposure, breathing exercises, and humidifiers cannot be transferred directly to this product.

A future staged program would begin with non-emitting form models to test grip, reach, mouthpiece alignment, and social interpretation. A benchtop sealed-flow rig could then verify valve direction, backflow prevention, outlet occlusion, condensation capture, and cleanability without human inhalation. Only after exposure characterization and independent safety review should any ambient mist testing be considered. Behavioral evaluation would separate immediate perceived completion, craving, substitution behavior, and sustained smoking outcomes rather than treating them as interchangeable.

Conclusion

Pause with You reframes the smoking pause as a multisensory interaction rather than a nicotine-delivery object alone. The literature supports attention to sensorimotor behavior and conditioned cues, but it also warns that resemblance can trigger the very behavior a substitute seeks to change. The resulting concept preserves grip, oral ritual, breath cycles, and visible completion while transforming form and separating exposure pathways. Its value as a portfolio project lies in that reasoning: personal experience becomes a question; evidence becomes requirements; contradiction becomes a design decision; and uncertainty remains visible rather than being converted into an unsupported health claim.

References

- Caponnetto, P., et al. (2011). The efficacy and safety of an electronic nicotine-free inhalator for smoking cessation: A randomized controlled trial. *European Respiratory Journal*.
<https://pubmed.ncbi.nlm.nih.gov/21565921/>
- Centers for Disease Control and Prevention. Why quitting smoking is hard.
<https://www.cdc.gov/tobacco/campaign/tips/quit-smoking/quit-smoking-medications/why-quitting-smoking-is-hard/>
- Centers for Disease Control and Prevention. Preventing waterborne germs at home.
<https://www.cdc.gov/drinking-water/prevention/preventing-waterborne-germs-at-home.html>

- Donny, E. C., et al. (2007). Smoking without nicotine: Behavioral and physiological effects. *Psychopharmacology*.
<https://pubmed.ncbi.nlm.nih.gov/17222288/>
- U.S. Environmental Protection Agency. Use and care of home humidifiers. <https://www.epa.gov/indoor-air-quality-iaq/use-and-care-home-humidifiers>
- U.S. Food and Drug Administration. FDA-approved and FDA-cleared cessation products can help. <https://www.fda.gov/consumers/consumer-updates/want-quit-smoking-fda-approved-and-fda-cleared-cessation-products-can-help>
- Veilleux, J. C., & Skinner, K. D. (2015). Smoking, food, and alcohol cues on subsequent behavior: A qualitative systematic review. *Clinical Psychology Review*. <https://pubmed.ncbi.nlm.nih.gov/25618485/>
- World Health Organization. (2024). WHO clinical treatment guideline for tobacco cessation in adults. <https://www.who.int/publications/b/74755>
- Multisensory smoking cue reactivity study. <https://pubmed.ncbi.nlm.nih.gov/26475784/>
- Breathing interventions for stress and mental health: systematic review. <https://pubmed.ncbi.nlm.nih.gov/38137060/>
- Systematic review of ultrasonic humidifier emissions. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9828259/>